

3. Trend analysis : Analyze the time data associated with social media and analyse its trends.

```
import pandas as pd
import matplotlib.pyplot as plt
```

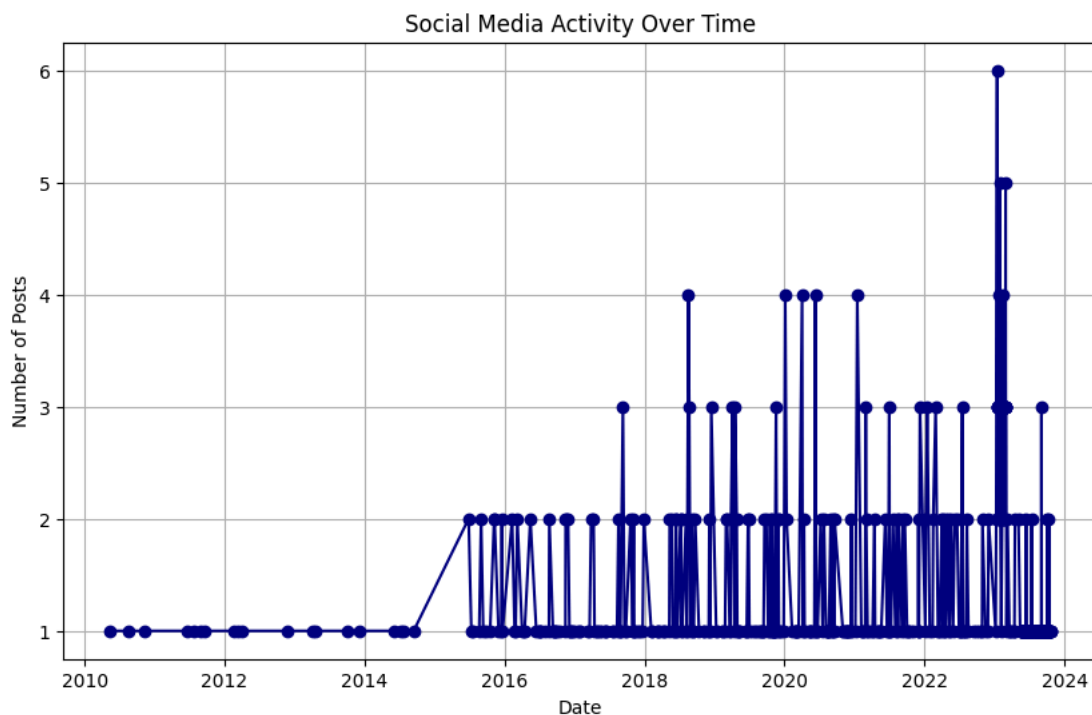
```
df = pd.read_csv('location.csv')
df.head()
```

	Unnamed: 0.1	Unnamed: 0	Text	Sentiment	Timestamp	User	Platform	Hashtags	Retweets	Likes	Country	Year	Month	Day	H
0	0	0	Enjoying a beautiful day at the park! ...	Positive	2023-01-15 12:30:00	User123	Twitter	#Nature #Park	15.0	30.0	USA	2023	1	15	
1	1	1	Traffic was terrible this morning. ...	Negative	2023-01-15 08:45:00	CommuterX	Twitter	#Traffic #Morning	5.0	10.0	Canada	2023	1	15	
2	2	2	Just finished an amazing workout! 👏 ...	Positive	2023-01-15 15:45:00	FitnessFan	Instagram	#Fitness #Workout	20.0	40.0	USA	2023	1	15	
3	3	3	Excited about the upcoming weekend	Positive	2023-01-15	AdventureX	Facebook	#Travel #Adventure	8.0	15.0	UK	2023	1	15	

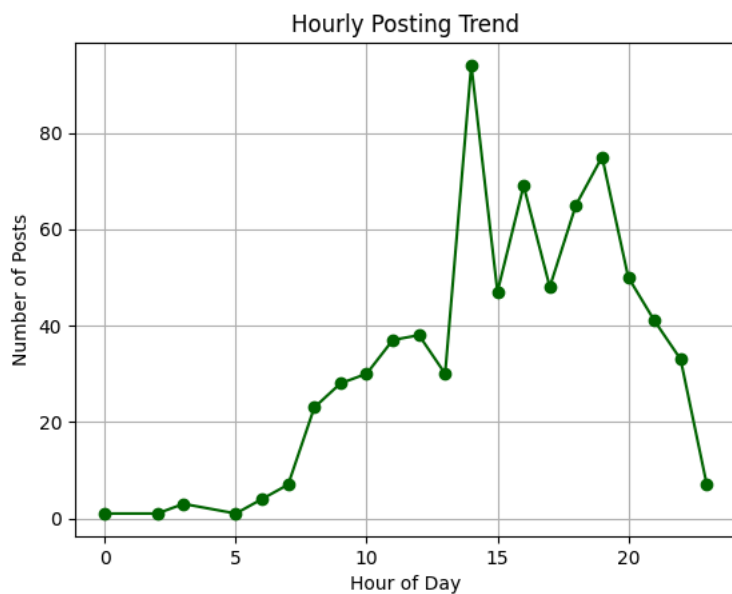
Next steps: [Generate code with df](#) [View recommended plots](#) [New interactive sheet](#)

```
df['Timestamp'] = pd.to_datetime(df['Timestamp'])
df['Date'] = df['Timestamp'].dt.date
df['Hour'] = df['Timestamp'].dt.hour
df['Month'] = df['Timestamp'].dt.month
df['Day'] = df['Timestamp'].dt.day
df['Year'] = df['Timestamp'].dt.year
```

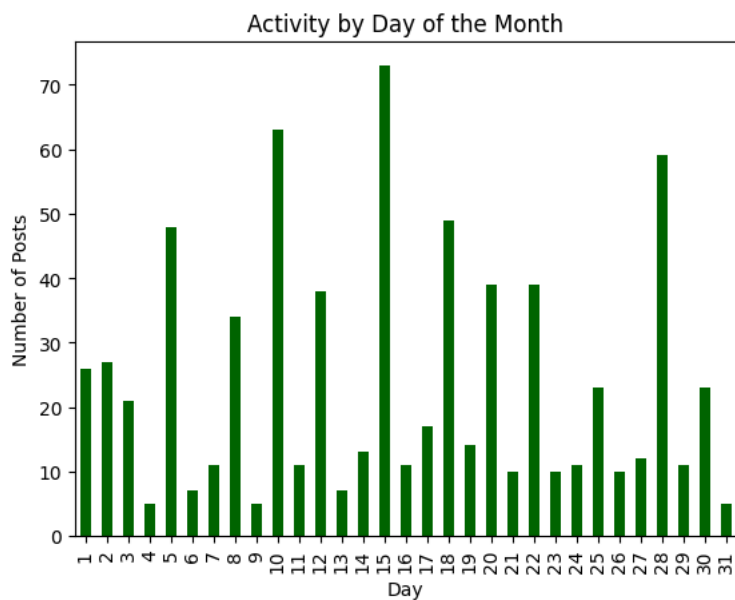
```
trend_by_date = df.groupby('Date').size()
trend_by_date.plot(figsize=(10, 6), marker='o', color='navy')
plt.title("Social Media Activity Over Time")
plt.xlabel("Date")
plt.ylabel("Number of Posts")
plt.grid(True)
plt.show()
```



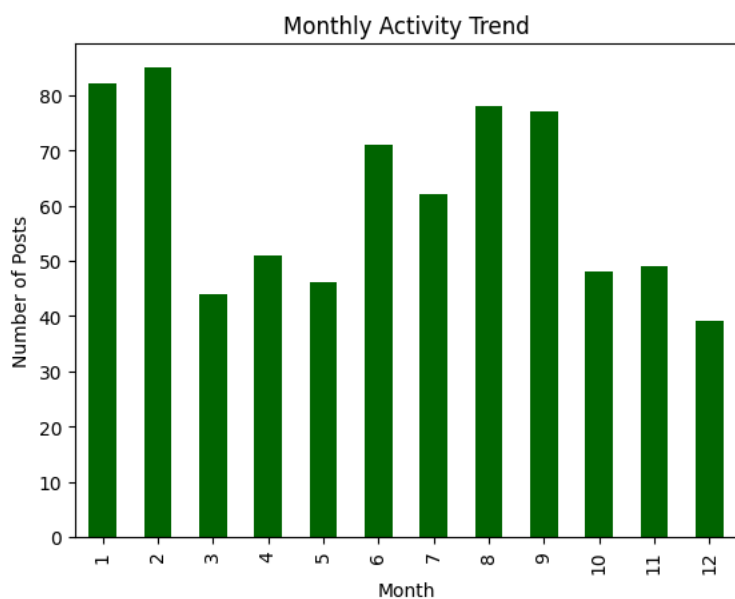
```
hourly_trend = df['Hour'].value_counts().sort_index()
hourly_trend.plot(kind='line', marker='o', color='darkgreen')
plt.title("Hourly Posting Trend")
plt.xlabel("Hour of Day")
plt.ylabel("Number of Posts")
plt.grid(True)
plt.show()
```



```
daily_trend = df['Day'].value_counts().sort_index()
daily_trend.plot(kind='bar', color='darkgreen')
plt.title("Activity by Day of the Month")
plt.xlabel("Day")
plt.ylabel("Number of Posts")
plt.show()
```



```
monthly_trend = df['Month'].value_counts().sort_index()
monthly_trend.plot(kind='bar', color='darkgreen')
plt.title("Monthly Activity Trend")
plt.xlabel("Month")
plt.ylabel("Number of Posts")
plt.show()
```



```
country_time_trend = df.groupby(['Date', 'Country']).size().unstack(fill_value=0)

country_time_trend.plot(figsize=(12, 6))
plt.title("Posting Trends Over Time by Country")
plt.xlabel("Date")
plt.ylabel("Number of Posts")
plt.legend(title='Country', bbox_to_anchor=(1.05, 1), loc='upper left')
plt.tight_layout()
plt.show()
```

